

RESEARCH PAPER



Additive and non-additive epigenetic signatures of natural hybridization between fish species with different mating systems

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ABSTRACT

Hybridization is a major source of evolutionary innovation. In plants, epigenetic mechanisms can help to stabilize hybrid genomes and contribute to reproductive isolation, but the relationship between genetic and epigenetic changes in animal hybrids is unclear. We analysed the relationship between genetic background and methylation patterns in natural hybrids of two genetically divergent fish species with different mating systems, *Kryptolebias hermaphroditus* (self-fertilizing) and *K. ocellatus* (outcrossing). Co-existing parental species displayed highly distinct genetic (SNPs) and methylation patterns (37,000 differentially methylated cytosines). Hybrids had predominantly intermediate methylation patterns (88.5% of the sites) suggesting additive effects, as expected from hybridization between genetically distant species. The large number of differentially methylated cytosines between hybrids and parental species ($n = 5,800$) suggests that hybridization may play a role in increasing genetic and epigenetic variation. Although most of the observed epigenetic variation was additive and had a strong genetic component, we also found a small percentage of non-additive, potentially stochastic, methylation differences that might act as an evolutionary bet-hedging strategy and increase fitness under environmental instability.

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Introduction

Hybridization is a source of evolutionary novelty [1,2] and can promote speciation through the generation of genetic and phenotypic variation, which could precede adaptive radiation [3,4]. Hybridization can result in heterosis or hybrid vigour, with hybrids displaying higher performance than the parents [5], but also in hybrid incompatibility resulting in loss of fitness, increased mortality, and increased reproductive isolation between parental species [6]. Interspecific hybridization rates are particularly high in plants (40%) [7] and seems to be a phenomenon uniformly distributed across animals [8]. Hybrid incompatibility, encompassing hybrid inviability and sterility is, together with reduced hybrid fitness, one of the strongest

postzygotic mechanisms acting as isolating barriers for interspecific hybridization [9]. Hybrid incompatibility can be caused by interactions between incompatible parental genetic alleles [10,11], changes in regulatory elements [12], transposable element activity [13,14] or chromosomal rearrangements [15], and also by cytosine methylation [16,17] and changes in gene expression patterns [18,19]. For instance, a single epigenetically inactivated gene (HISN6B) is responsible for hybrid incompatibility between strains of *Arabidopsis thaliana* [6].

Interactions between parental genomes can result in additive or non-additive gene expression patterns [20]. For example, hybrids between farmed and wild Atlantic salmon or between recently-diverged pupfish species have shown

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